AI/HPC Lightning Talks



OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

Enabling Petawatt AI Clusters with Real-time Energy Routing (RER)

Gerald Bode & Florian Engel

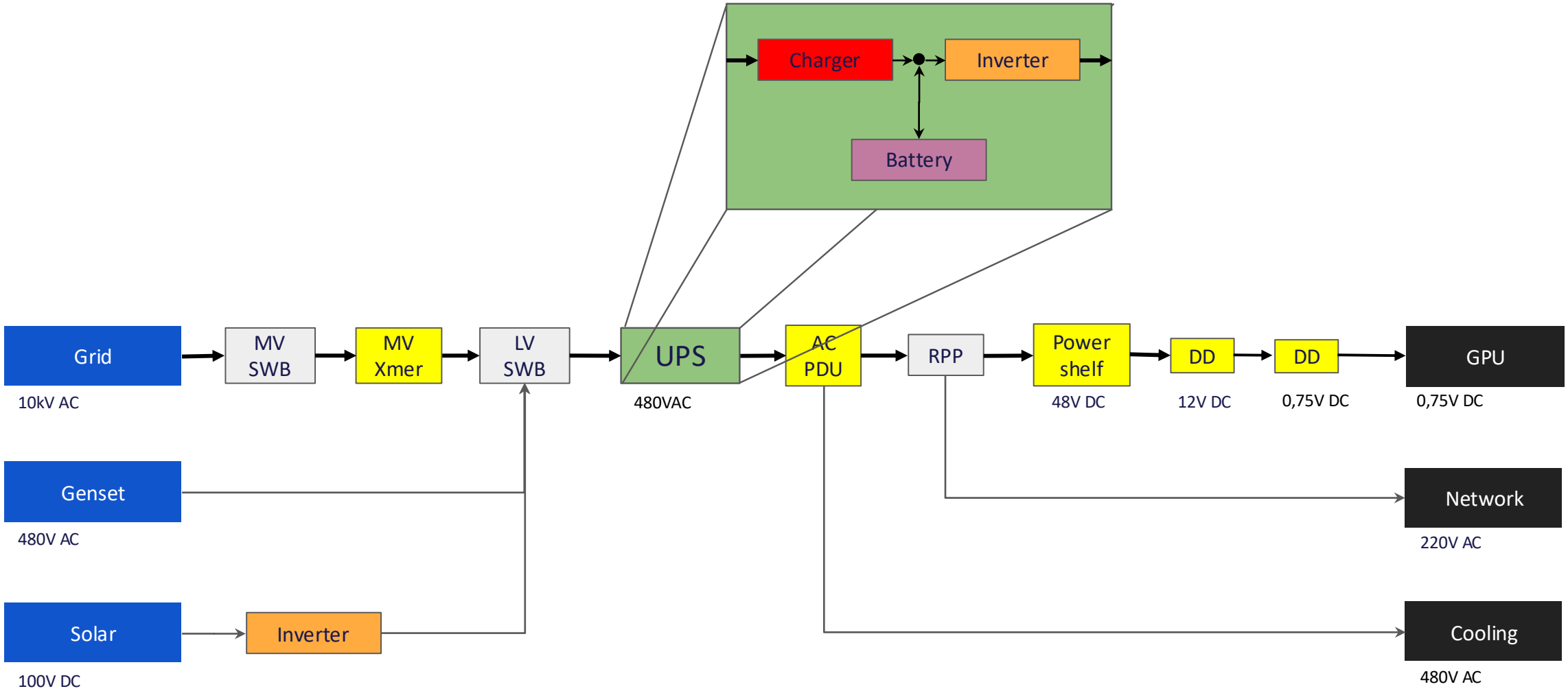
Nlevel



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Current Technology



Enabling Petawatt AI Clusters together

The Power Wall

Monolithic AC-UPS-DC chains cannot scale.

Microsecond transients crash centralized hardware.

The "Analog Tax" is wasting your grid capacity.

Real-Time Energy Routing (RER)

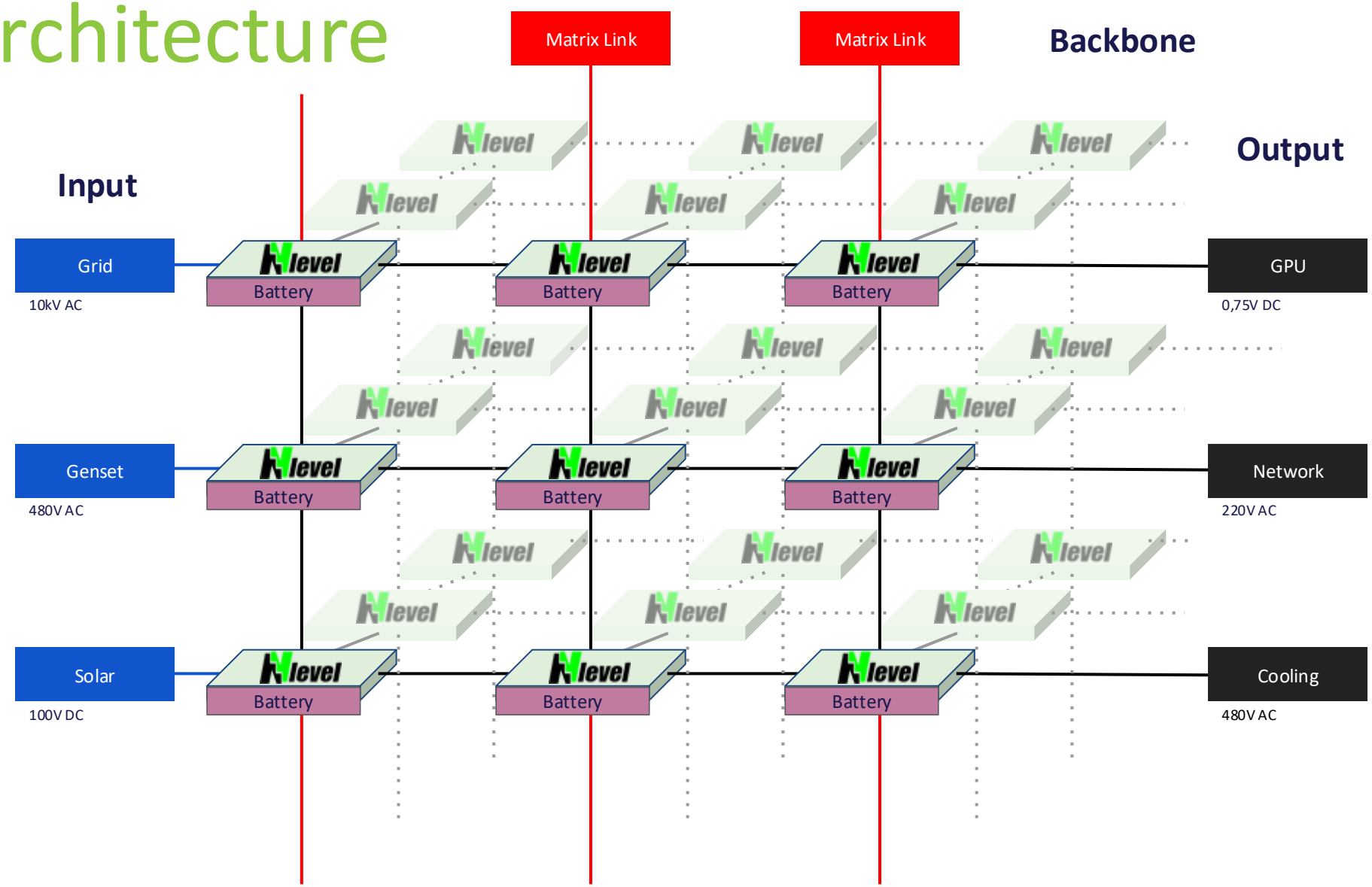
Energy is not converted or reconstructed.

RER is completely agnostic to voltage and power (AC or DC).

Software-Defined Matrix: Energy is routed in addressable packets directly to the silicon.

Zero Central Failure:
Masterless, self-healing swarm matrix buffers di/dt transients at the rack.

RER – Architecture



Let’s create an OCP Workstream for Energy Routing

OSI Layers - for Realtime Energy Routing

- worldwide frame format
- worldwide paket format
- Energy PHY
- realtime energy routing module Hot-plug
- mechanical interfaces power / data plugs
- global time synchronization
- automatic link aggregation

Let's Define the Interface Together

- We are opening the Real-Time Energy Routing (RER) specification to the community.

- By co-designing the hardware and software interfaces, we ensure that all Datacenter components — regardless of the vendor — can connect, speak the same language, and route energy seamlessly.

One Standard.

Any Vendor.

Infinite Scaling.

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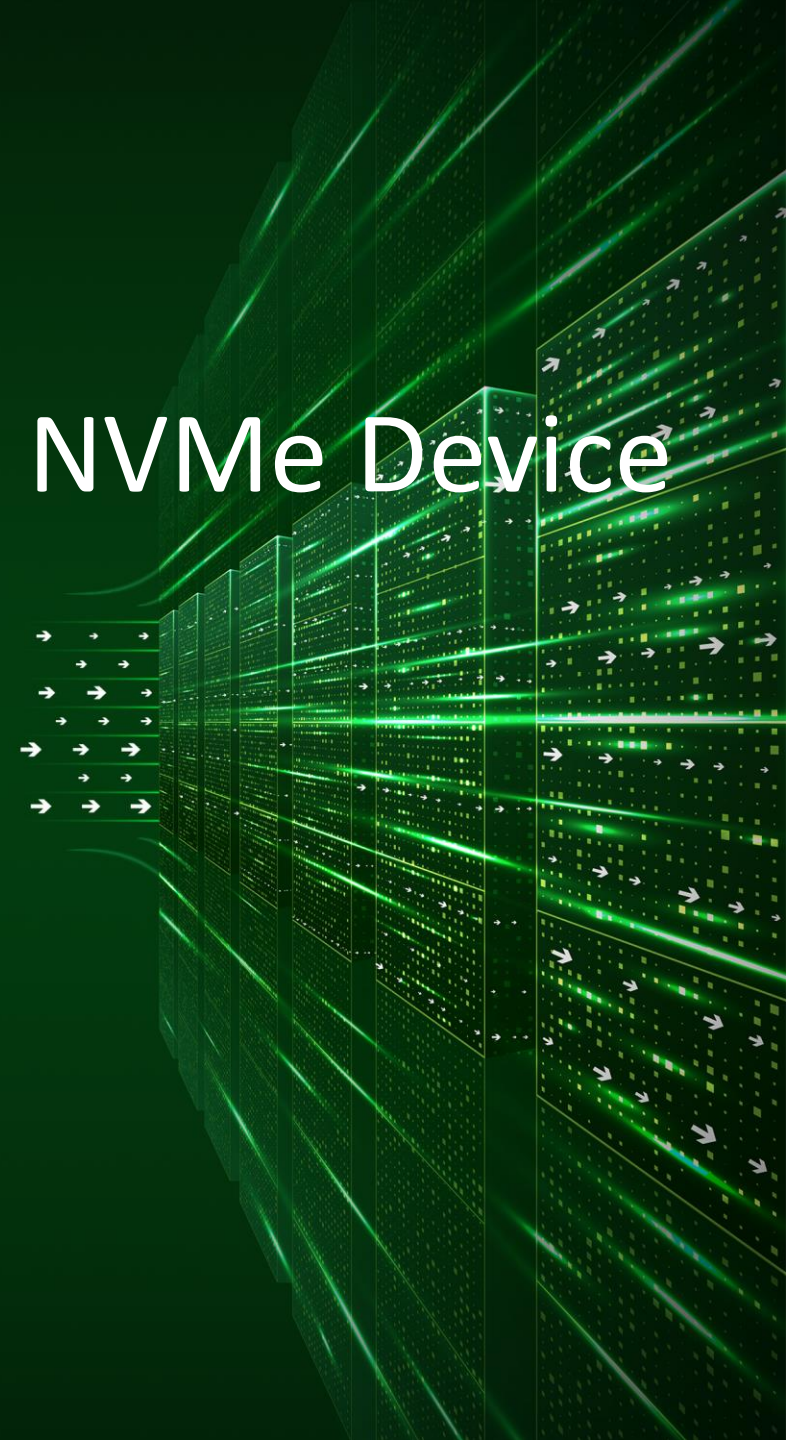
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NVMeNext: Rearchitecting a Virtual NVMe Device for GPU-centric Storage Systems

Injae Kang and Jaehoon Shim

Ph.D. Student, Seoul National University



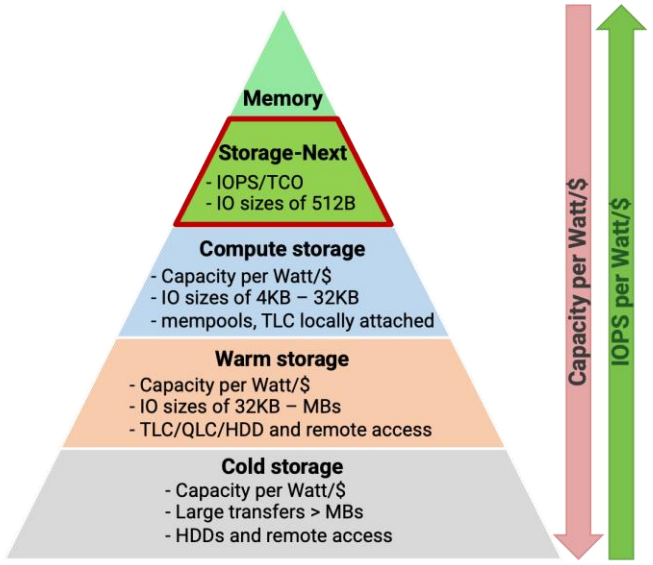
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SSD Emulator for GPU-Initiated I/O

- Massive memory footprints in modern GPU workloads
  - Recent systems offload datasets from HBM to the storage tier
  - CPU involvement in GPU-storage data transfers becomes a bottleneck
  - In GPU-initiated I/O, GPUs directly issue I/Os without CPU intervention

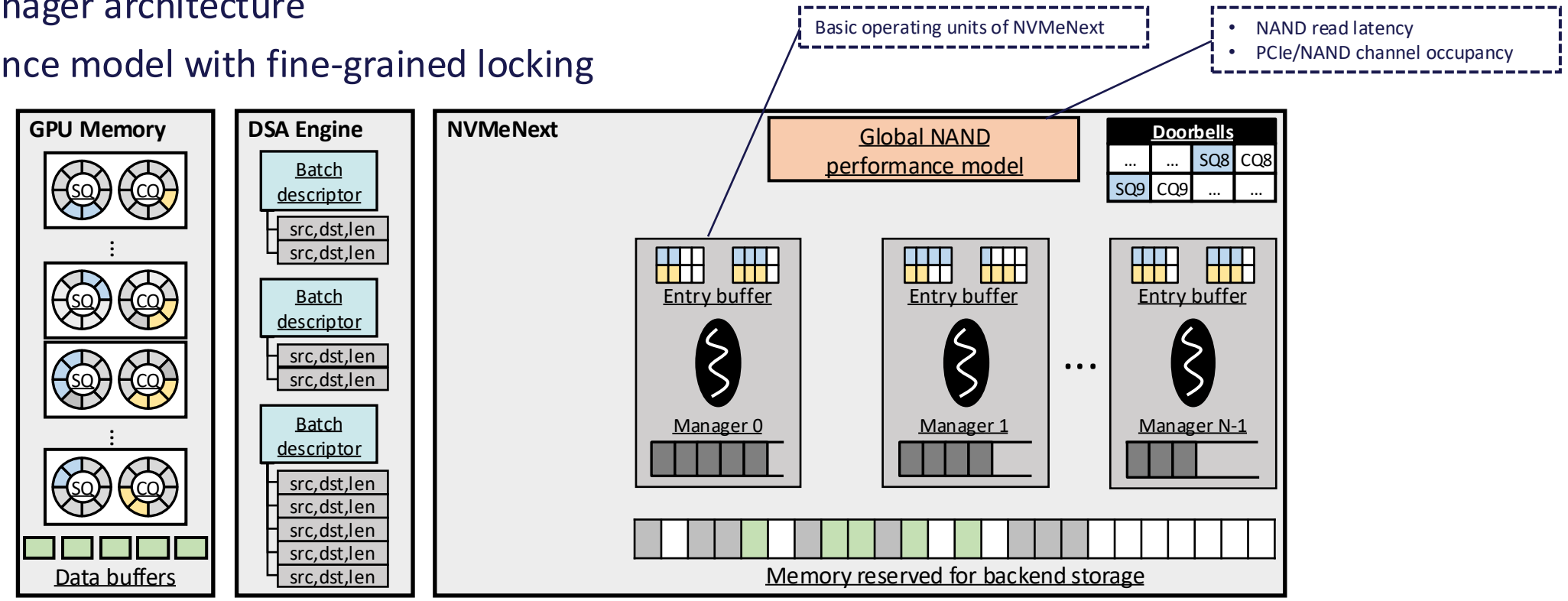
- NVIDIA's Storage-Next vision
  - Storage as an extension of GPU memory
  - SSD emulators must scale to MIOPS-class throughput and support efficient P2P with GPUs



"Advancing Memory and Storage Architectures for Next-Gen AI Workloads", FMS'25

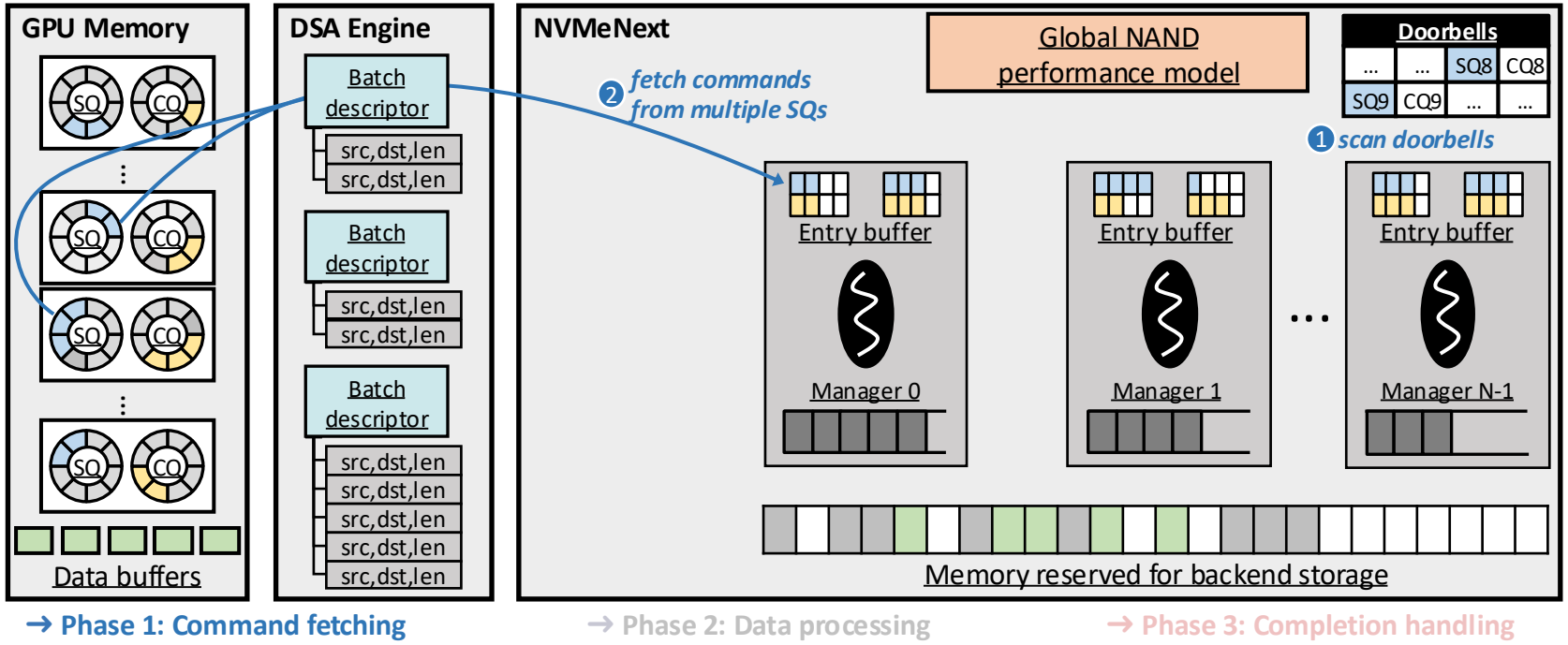
NVMeNext Overview

- Rearchitected NVMeVirt[FAST '23] for GPU-initiated I/O
  - Multi-manager architecture
  - Performance model with fine-grained locking



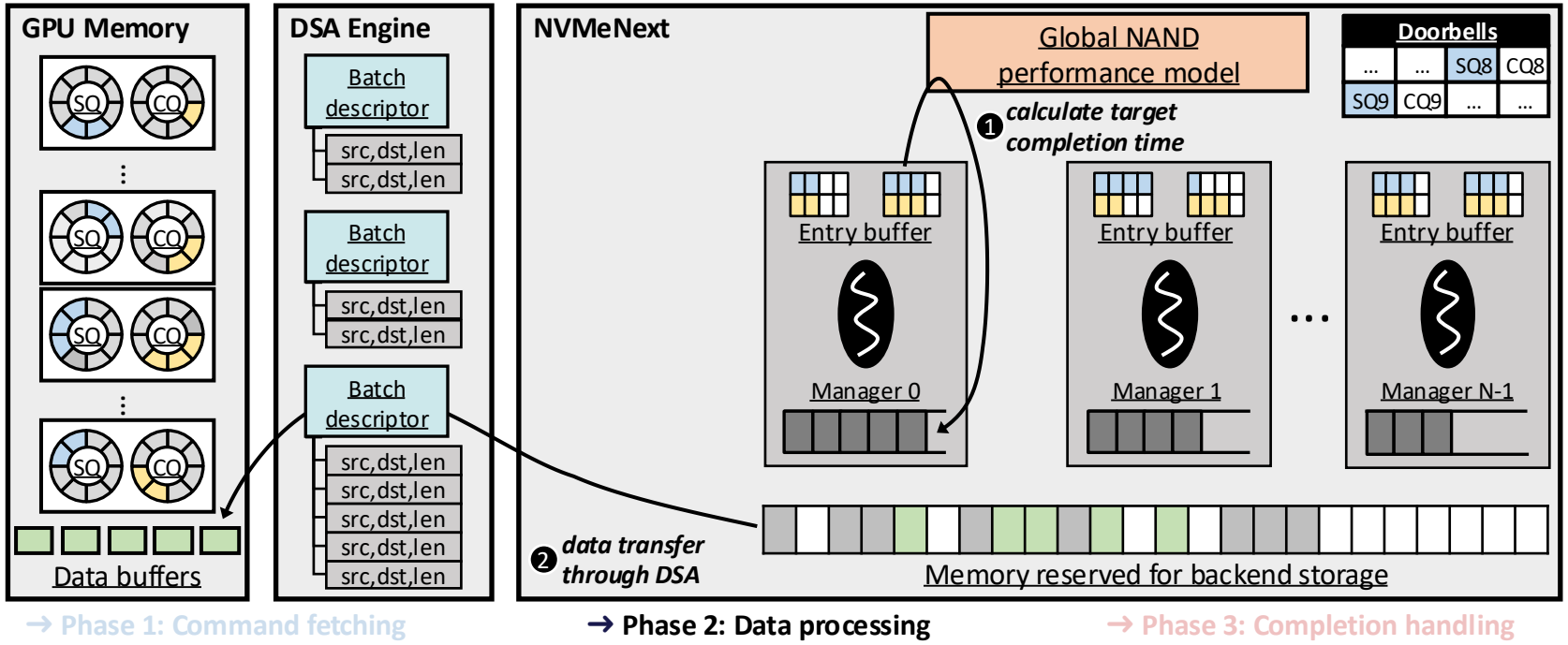
NVMeNext Workflow

- Phase 1: Command fetching



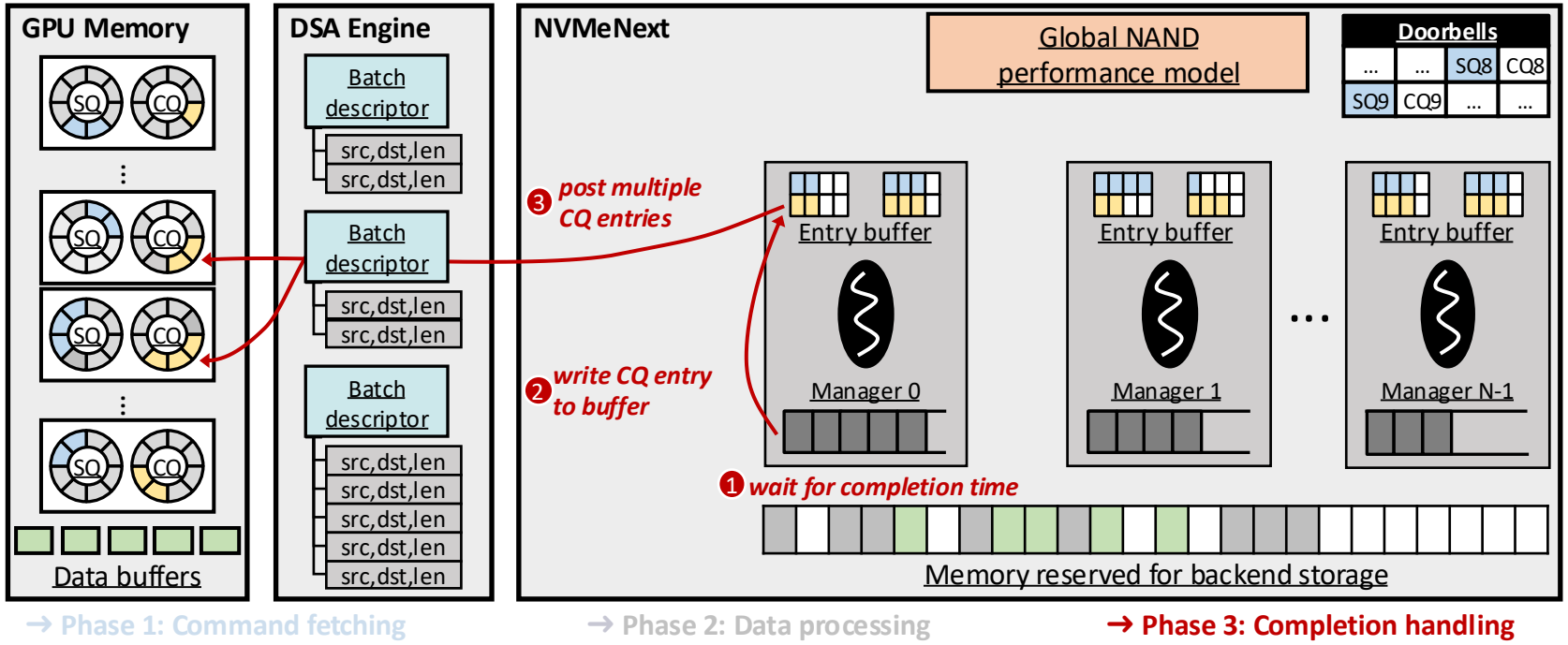
NVMeNext Workflow

- Phase 2: Data processing



NVMeNext Workflow

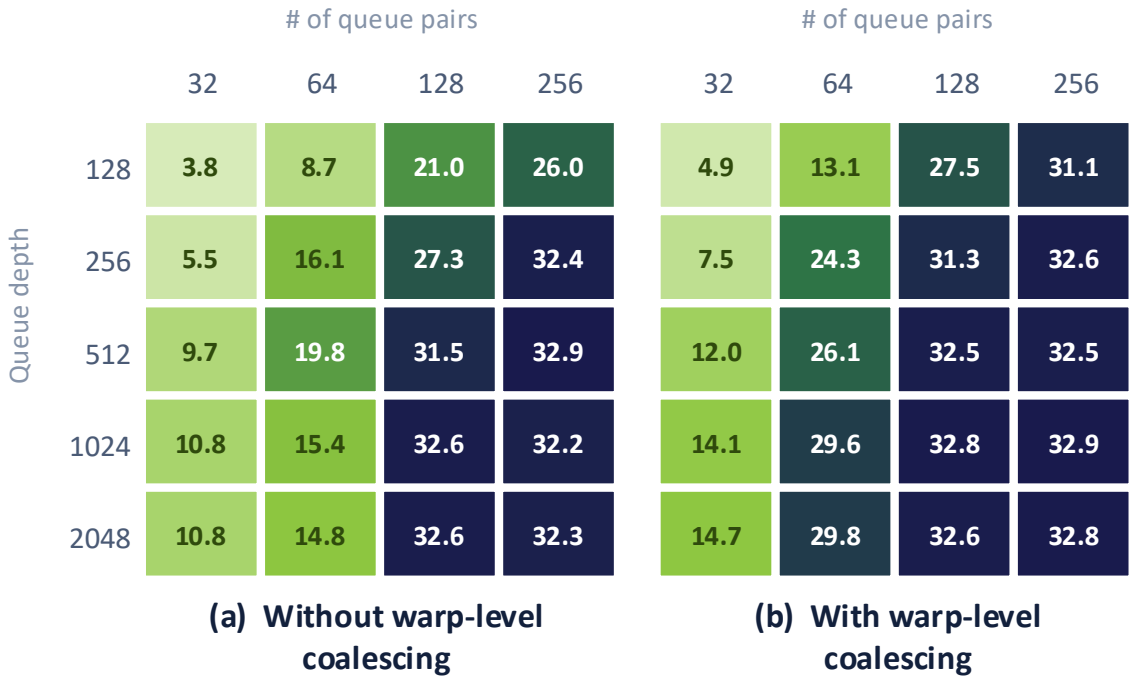
- Phase 3: Completion handling



System-Level Analysis with NVMeNext

- SSD configuration
  - 16 NAND channels, 32 planes/channel, and 15 µs NAND read latency

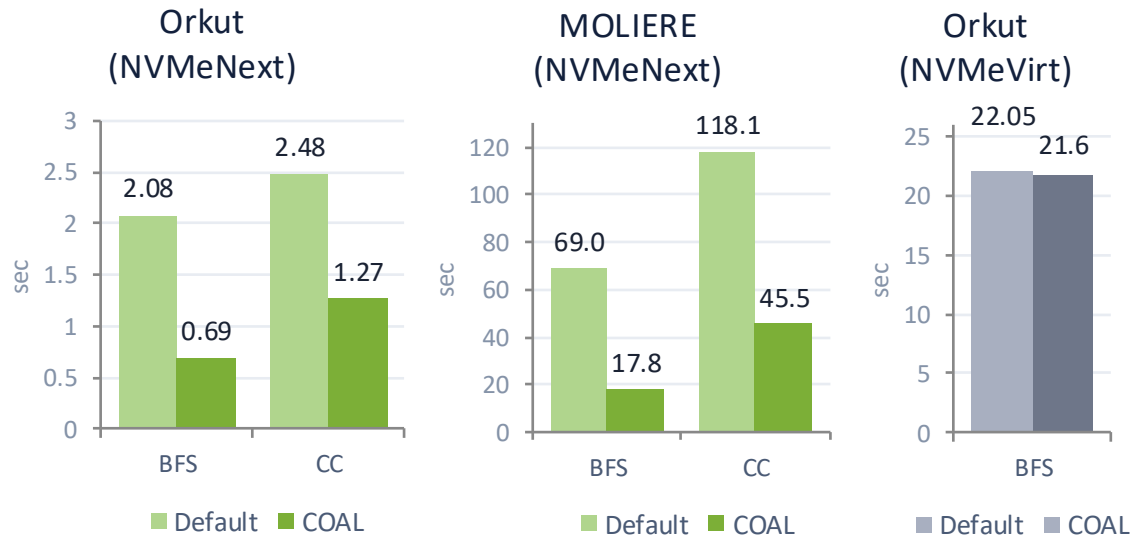
- Impact of NVMe queue pair topology



(a) Without warp-level coalescing

(b) With warp-level coalescing

- Graph analytics workloads



Conclusion

- **NVMeNext**: a scalable NVMe SSD emulator for GPU-centric storage systems

1. **Rearchitected for GPU-initiated I/O**

   Multi-manager design, NAND perf model with per-plane locking, and DSA-based batch processing

2. **MIOPS-scale request handling**

   Reaches the theoretical peak under real GPU-initiated workloads

3. **A tool for system-level analysis**

   Captures how SSD-side properties and GPU-side optimizations jointly shape performance

Try it!

Available on GitHub

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Millisecond-Scale Load Smoothing and Grid Contingency Ride-Through at GW-Scale: Future-Proofing for Emerging Grid Codes

M A Awal & Jason Barmann

EPC Power Corporation

Poway, CA, USA

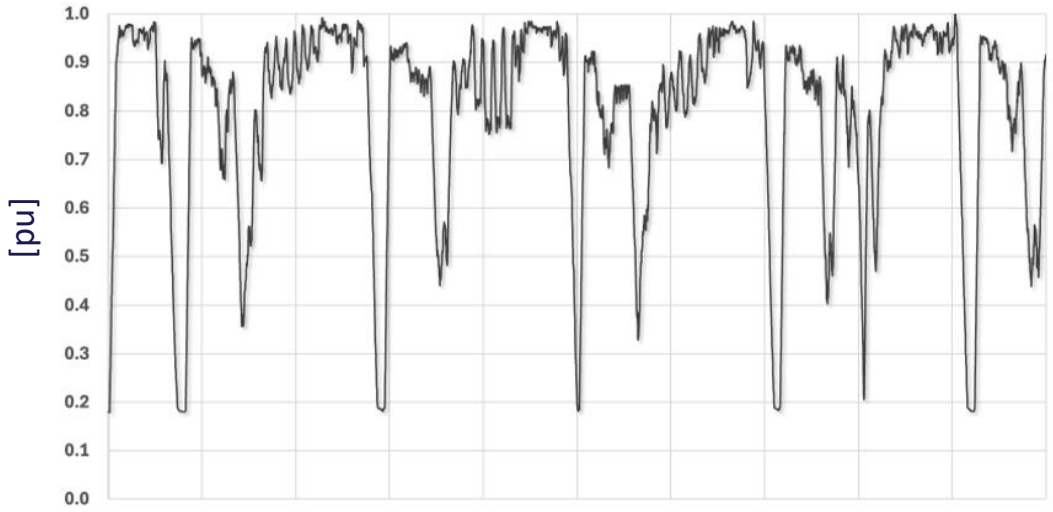


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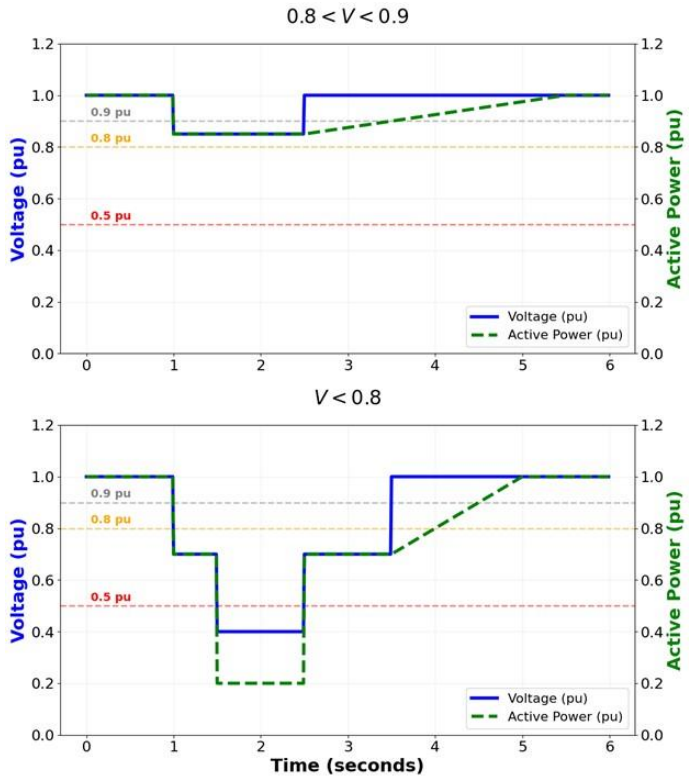
epcpower

Challenges at GW Scale



Representative Load Volatility

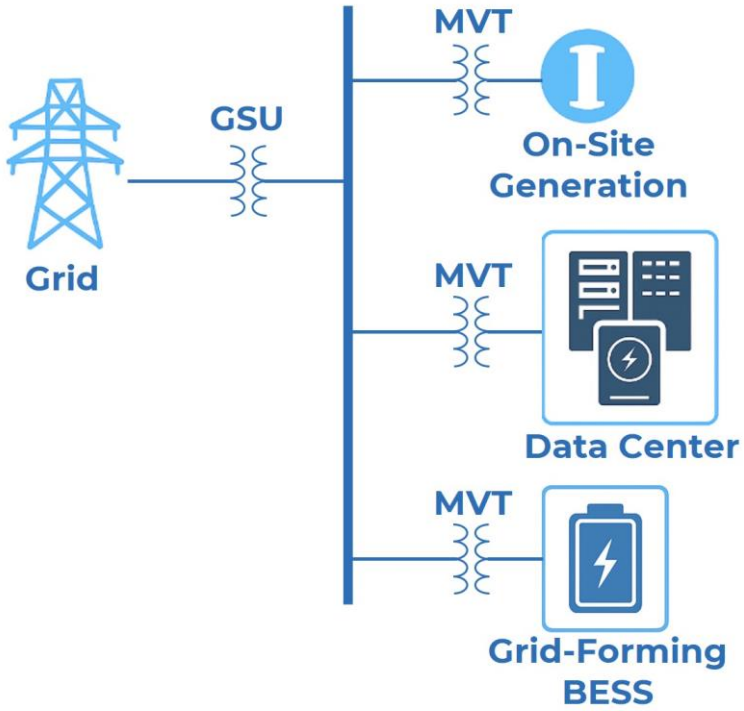
- Extreme load variability at unprecedented scale
  - Sub-synchronous oscillation/ torsional vibration of generator shaft → accelerated failure of on-site generation
- Emerging grid code mandates
  - Power ramp-rate limit
  - Ride-through requirements

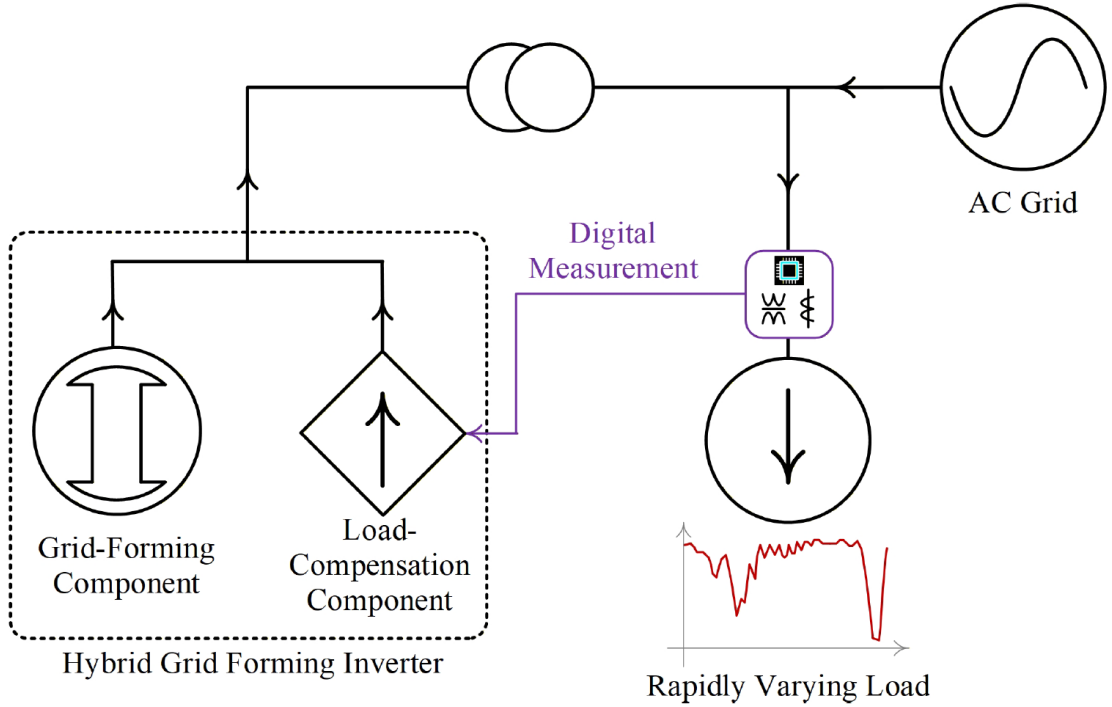


Low-Volt Ride-Through Requirement

ERCOT NOGRR282

Agile Grid-Forming™ BESS





- Non-invasive to proven infrastructure (on-site generation, distribution, UPS etc.)
- Single power-conversion stage connects parallel to load at same point-of-interconnection (POI)
- Double duty – load smoothing & large-load ride-though

- A single inverter operates as two components – grid forming & explicit load compensation component
- Direct measurement over Ethernet PHY (load or source side); fast compensation loop (<2ms)

Lab Validated Smoothing Performance

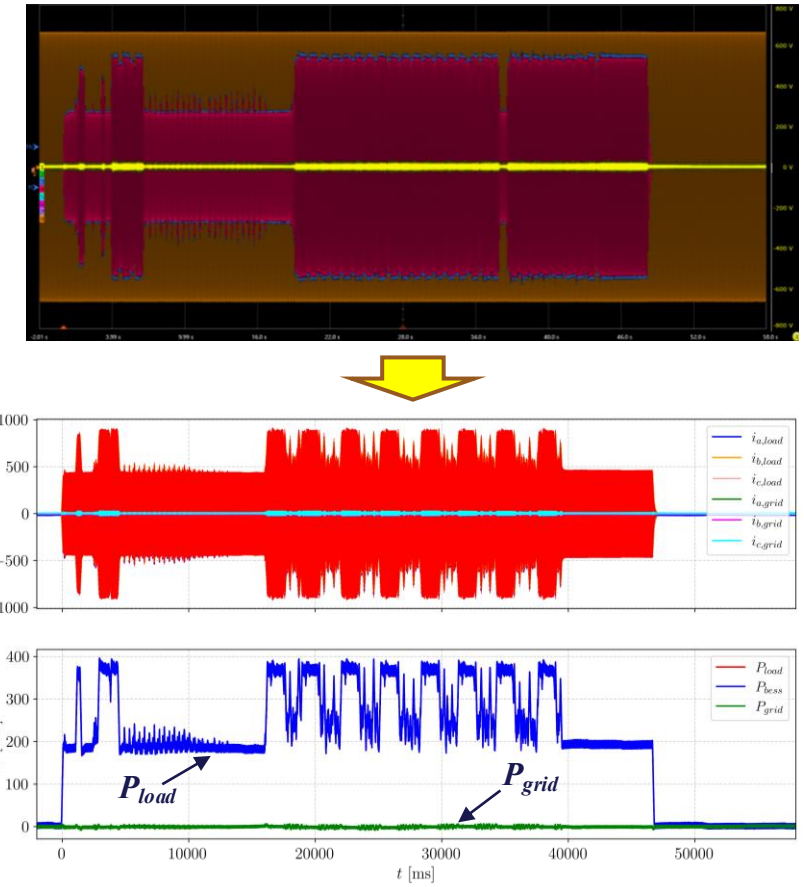


- Hardware test setup to demonstrate load-smoothing performance
- Programmable SCR via grid emulator to verify performance across various interconnection strengths

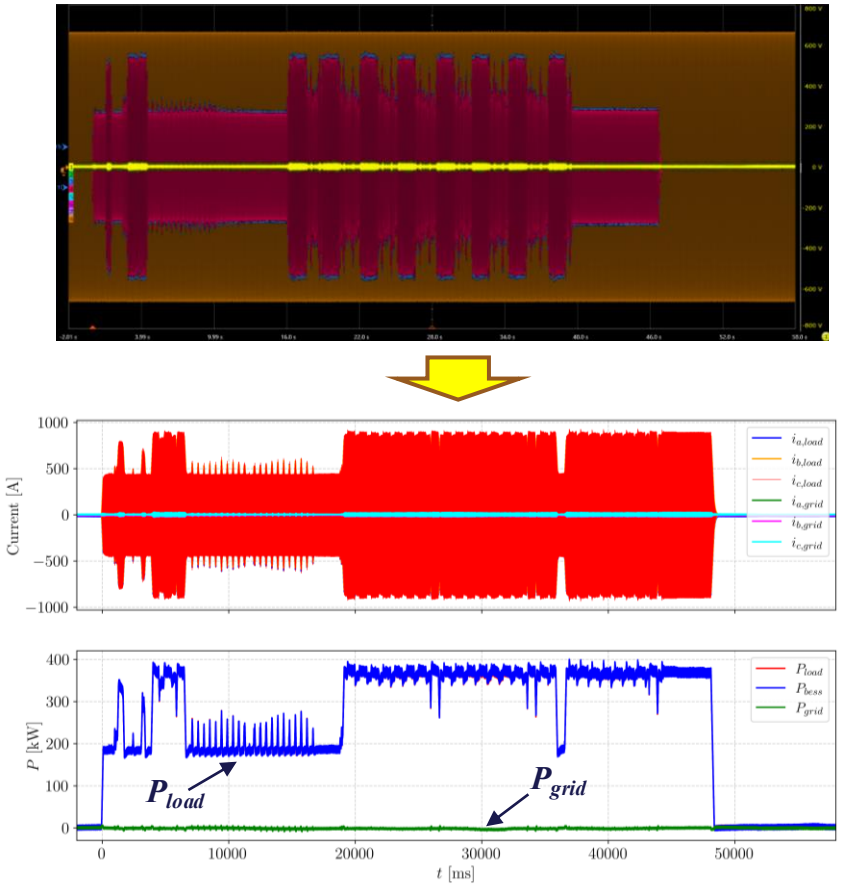


0 to 95% Load Step

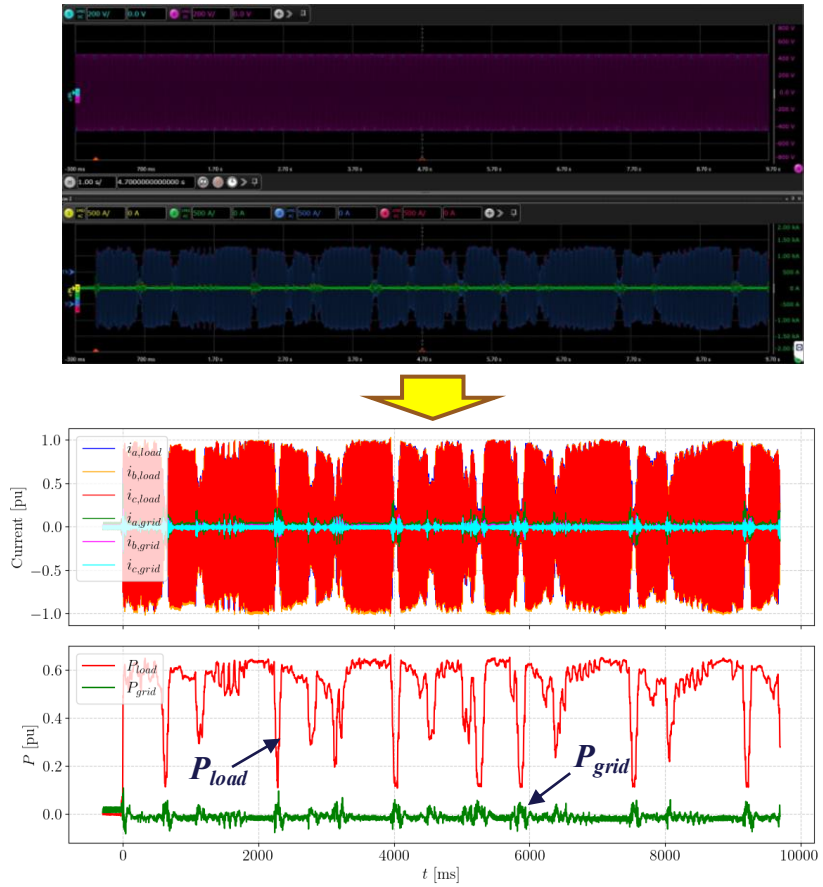
Lab Validated Smoothing Performance



Training Profile #1

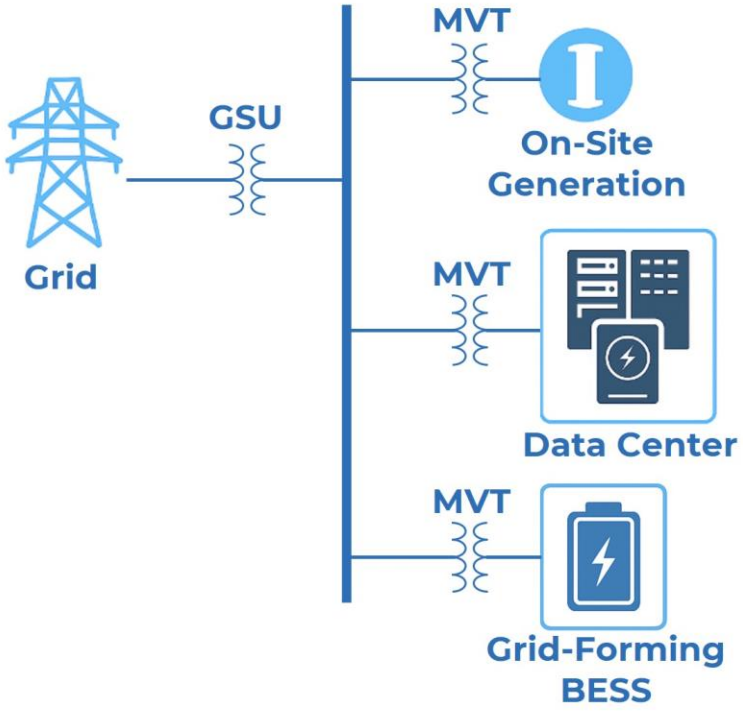


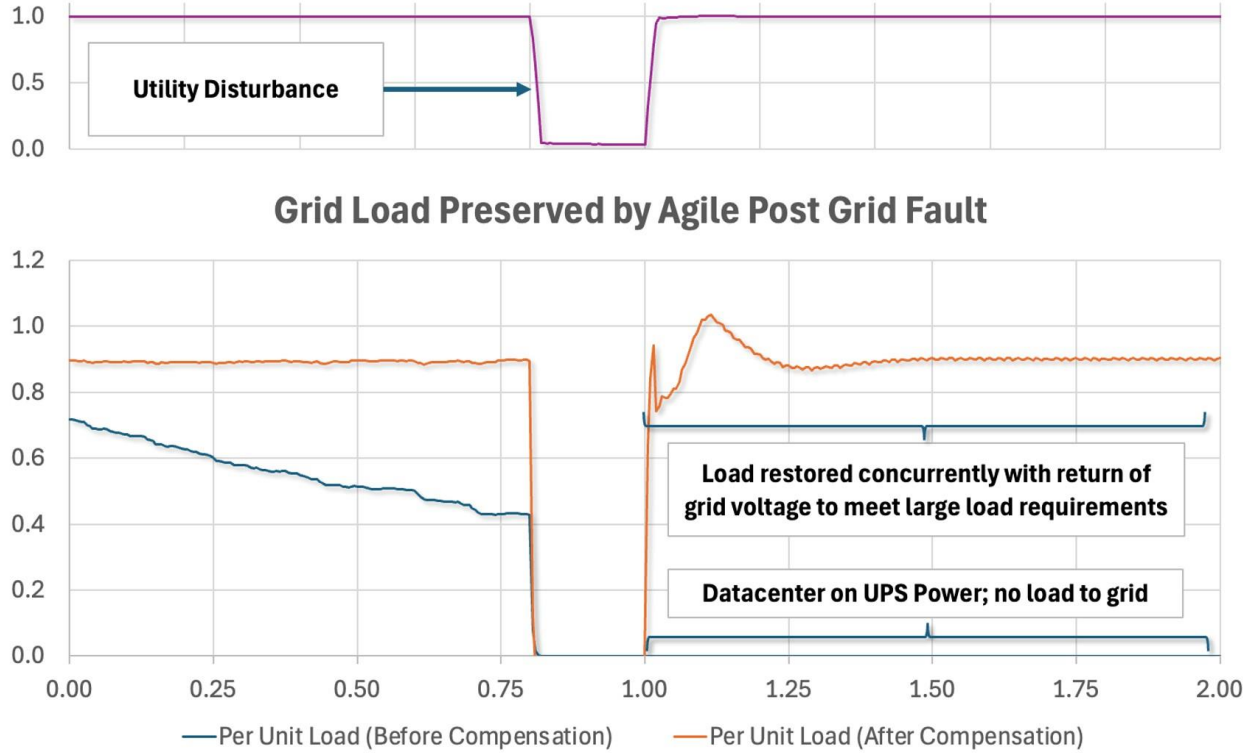
Training Profile #2



4x Accelerated Profile

Grid-Contingency Ride-Through





Retains pre-contingency load level immediately at fault clearance

One Architecture, Unit to GW-Scale



5 MW Building Blocks
Hundreds of blocks aggregate to multi-GW plants

Agile Measurement over Ethernet PHY
<2 ms compensation latency

Non-Invasive at the Point-of-Interconnection (POI)
Single power-conversion stage parallel-connected at POI

Proven Performance
Millisecond load smoothing + grid-contingency ride-through

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Single-Stage Vertical 48V-to-Core Power Delivery for Future AI Processors

Zhenshuai Rong, Yingchao Chi

University of Cambridge



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Why Single-Stage?

Data-center scale makes every efficiency point matter

Two-stage vs. single-stage architecture

48 V — IBC — 6/12 V intermediate bus — VRM — Core

extra conversion stage | intermediate-bus distribution loss

Remove the intermediate bus

48 V — Single-stage vertical VRM — Core

simplified power path | higher end-to-end efficiency potential

Global data-center electricity demand

485 → 950 TWh/year

2025 → 2030 projection

A ~2 percentage-point end-to-end efficiency gain

≈ 19 TWh/year

saved electricity

≈ $1.9B/year

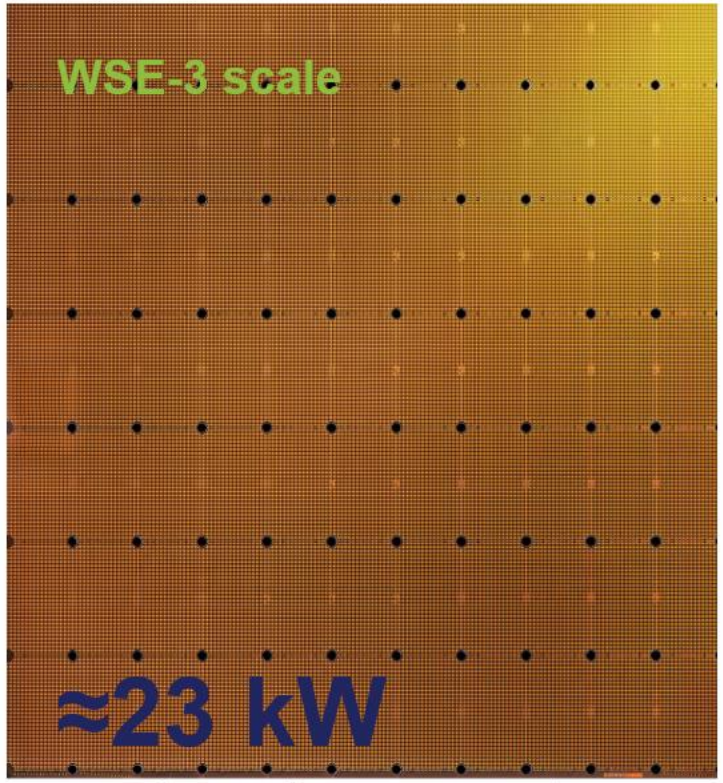
at $0.10/kWh

One fewer conversion stage means significant energy and cost savings at scale.

Source: IEA, Key Questions on Energy and AI (2026). Electricity-cost calculation is illustrative.

Why Low-Voltage Buses Do Not Scale?

High current, high I^2R loss, and limited energy storage at multi-kW scale



system-scale power example

6 V bus

≈ 3.8 kA

at 23 kW

→ **0.1 mΩ** distribution resistance

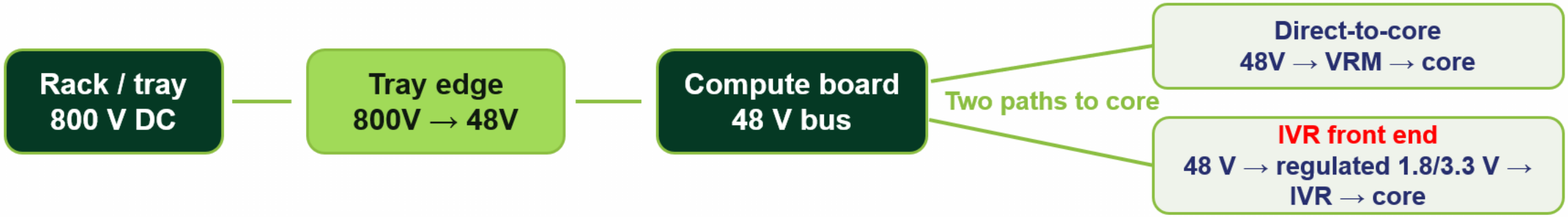
→ **≈1.47 kW** bus conduction loss

CURRENT	LOSS	ENERGY
6 V bus 8× current vs 48 V	I^2R penalty 64× bus loss vs 48 V	$E \propto CV^2$ 48 V stores 64× energy

At high power and large system scale, low-voltage intermediate buses significantly increase current, decoupling demand, and distribution loss.

Why 48 V at the Board Level?

A common platform for direct-to-core and IVR-based power delivery



Why IVR? Keep most distribution at a higher voltage, then regulate very close to the die.

Key benefits of IVR for future high-current AI processors

Lower package current I ↓
1.8 V → 0.8 V near die
≈ 2.25 × lower input current

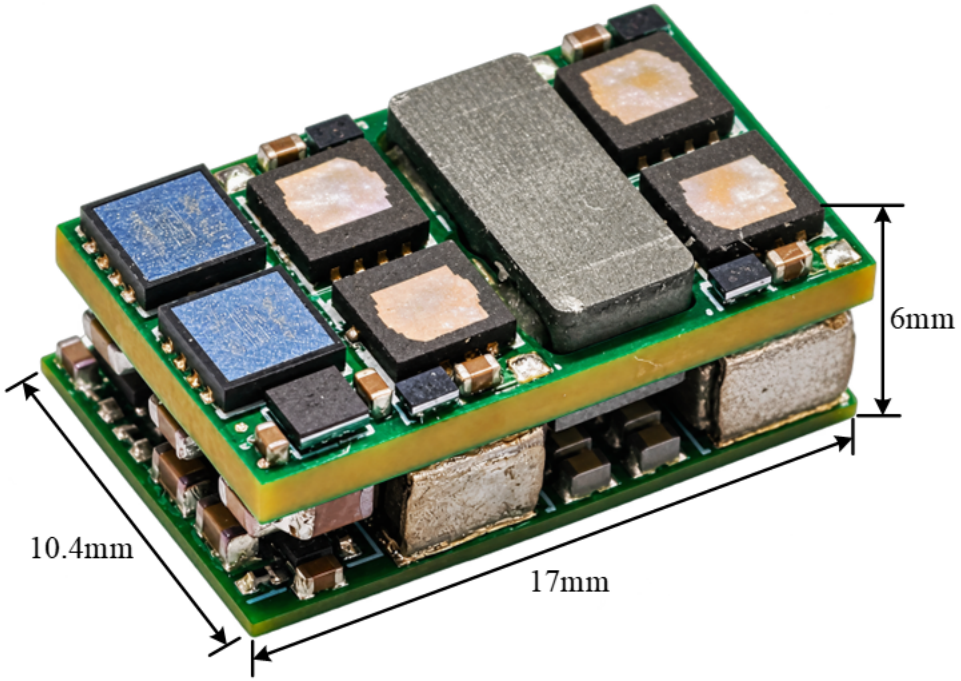
Less PDN stress R ↓
lower I^2R loss
reduced electromigration stress

Better transient margin L ↓
shorter inductive path
reduced voltage fluctuation

single-stage vertical 48V-to-core VRM can serve both direct-to-core and IVR-front-end roadmaps.

Proposed Solution Overview

Single-stage vertical VRM for multi-kiloampere processor power delivery



Regulated 54 V-to-core vertical VRM prototype

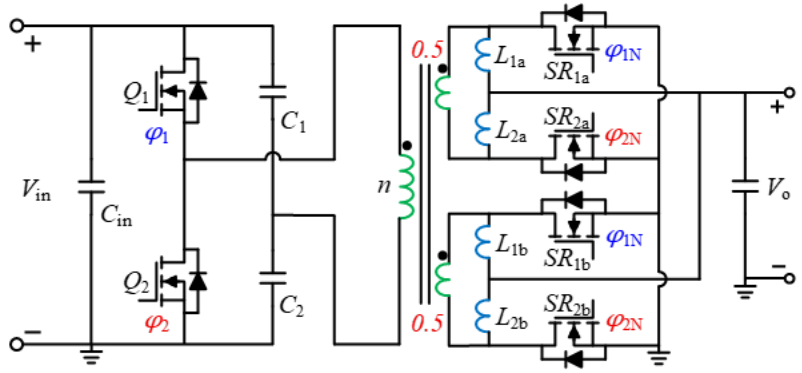
Core innovations

1. **Single-stage conversion**
   Fractional turns boost conversion ratio
2. **Vertical power delivery**
   Shorter high-current output path
3. **Direct top-side cooling**
   Short thermal path to the heatsink
4. **Modular scalability**
   Compatible with digital multiphase control

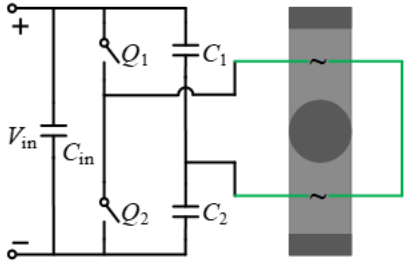
Vertical placement brings power conversion closer to the core

Fractional-Turn Current-Doubler Topology

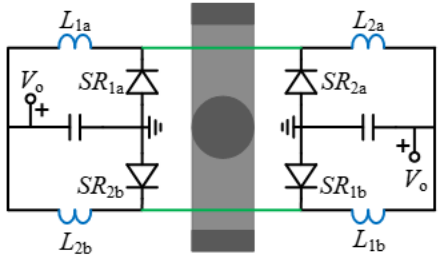
High conversion ratio + high output-current capability



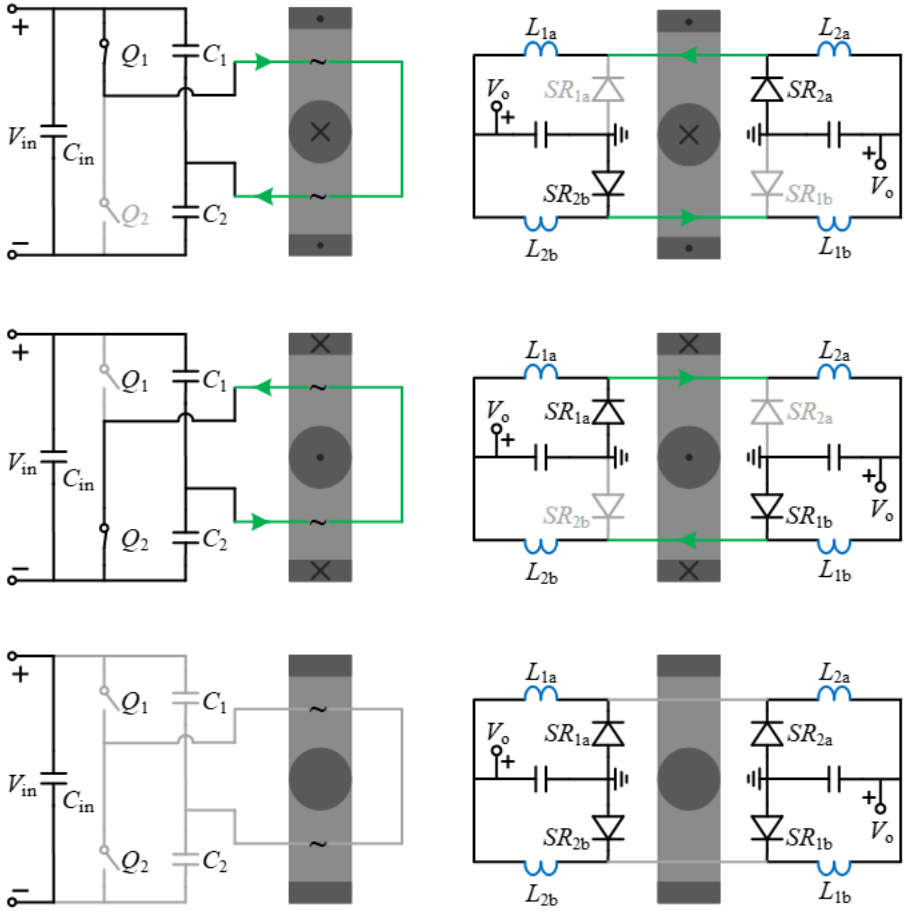
Half-bridge current doubler with fractional turns



Primary side



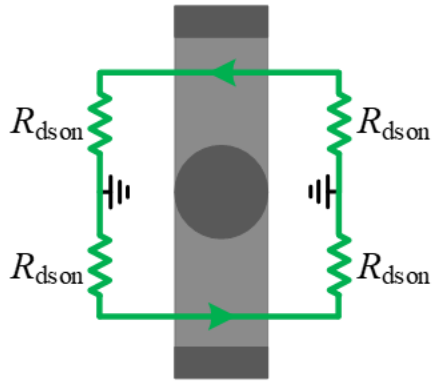
Secondary side



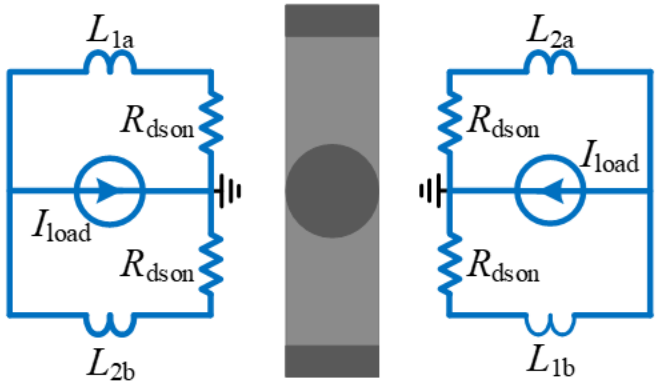
Operation modes

Current-Loop-Guided Architecture

Loop analysis → layout rule: planar AC circulation + vertical DC delivery



AC loop



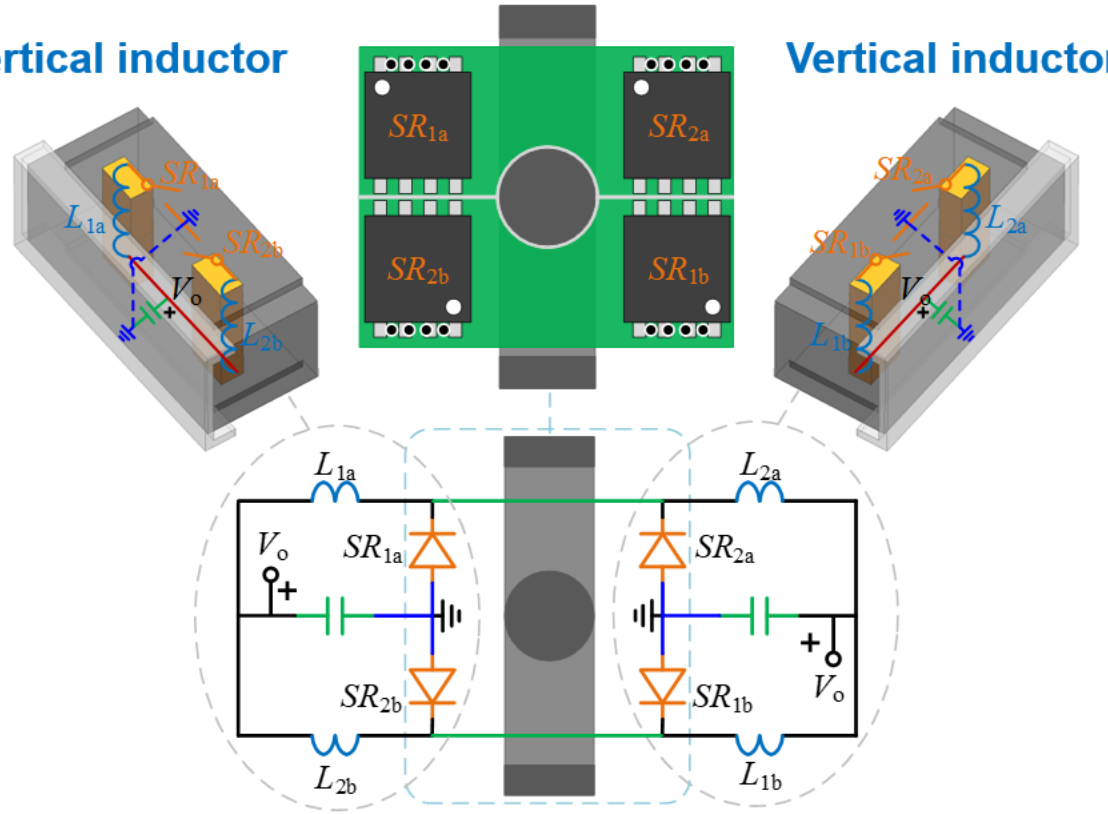
DC loop



Planar transformer

Vertical inductor

Vertical inductor



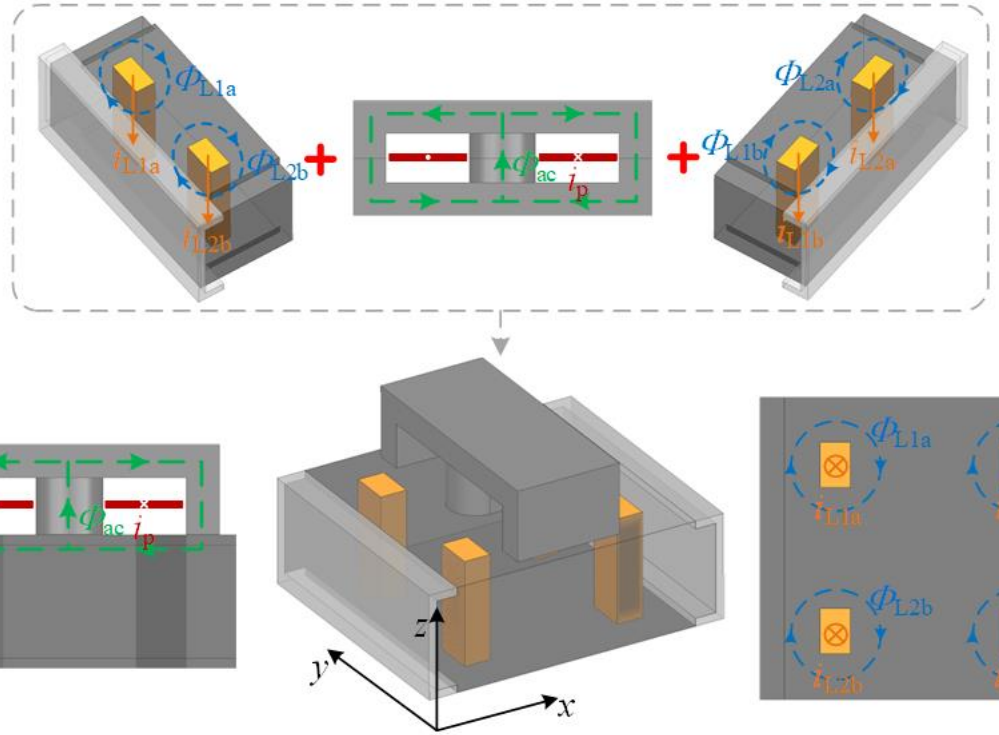
Ferrite core → high-frequency AC flux

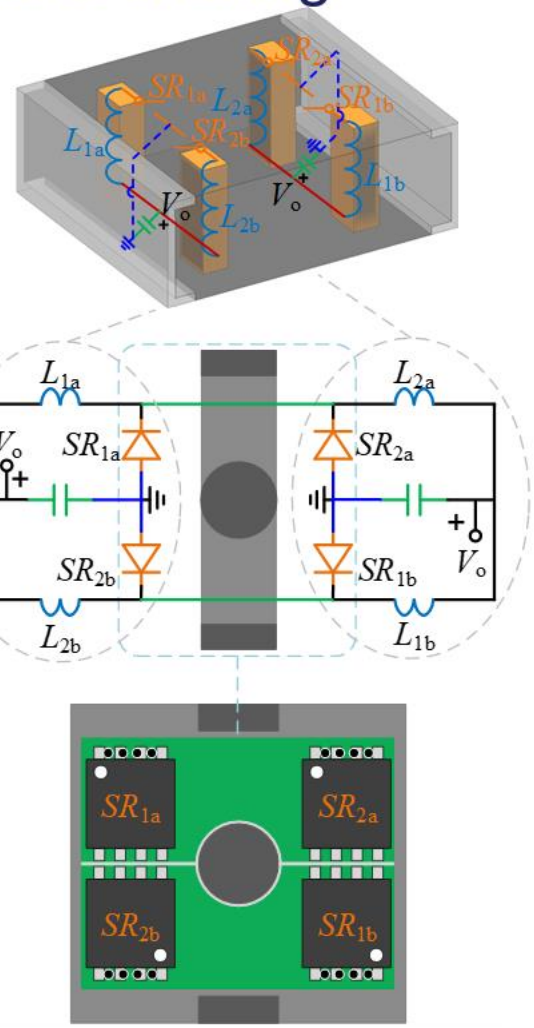
Powder core → DC-biased inductor flux

Magnetic Integration

Near-orthogonal flux paths enable transformer-inductor core sharing

Transformer flux ⊥ Inductor flux → Weak magnetic interaction





Side view

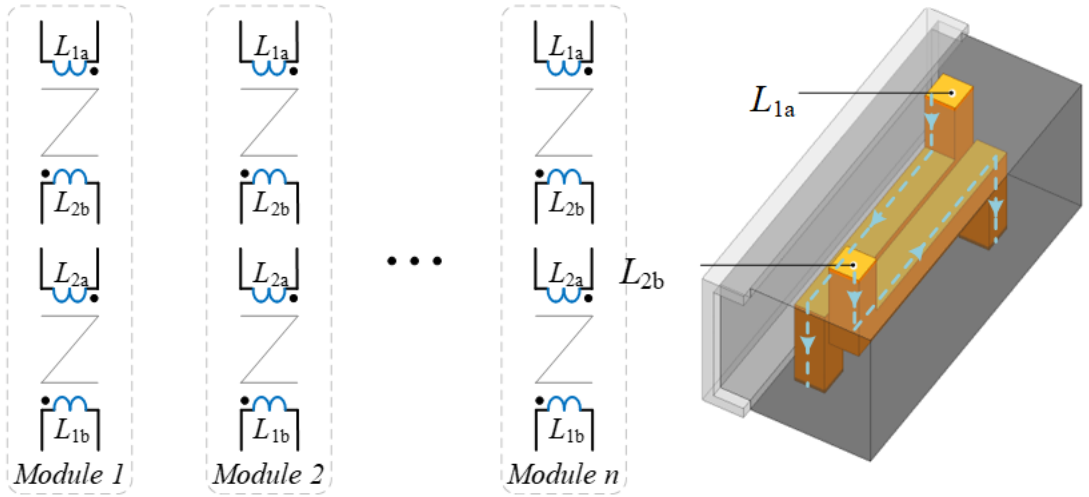
AC flux closes through ferrite + powder core

Top view

Inductor flux remains local in powder cores

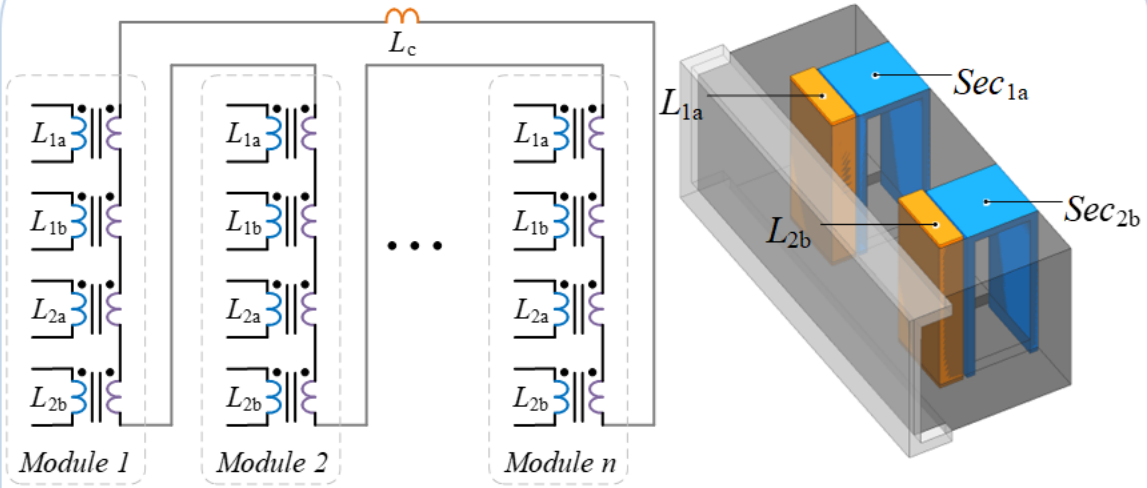
Transient Enhancement via Coupling

Negative coupling and TLVR enable faster transient response



Direct Negative Coupling

- Flux cancellation → higher current capability
- High steady-state *L* → low ripple
- Low transient *L* → fast response

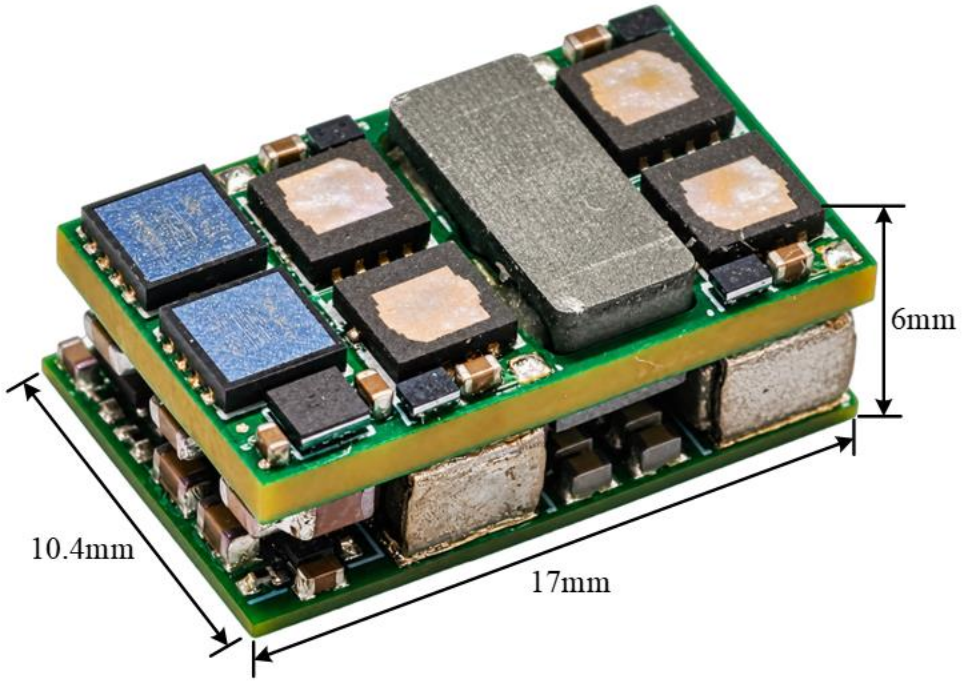


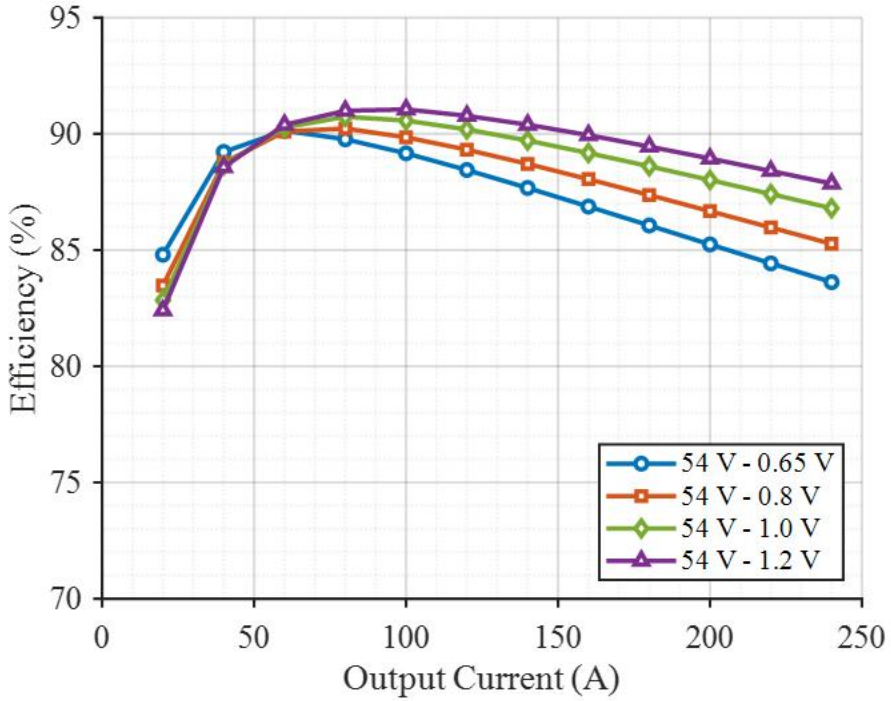
Indirect Coupling — TLVR

- Auxiliary-winding coupling → easy phase scaling
- Very low transient *L* → faster dynamics
- Leakage inductance as L_c → fewer components

Prototype & Measured Performance

Measured prototype validates high-current-density 54 V-to-core conversion





54 V → 0.8 V / 240A operation

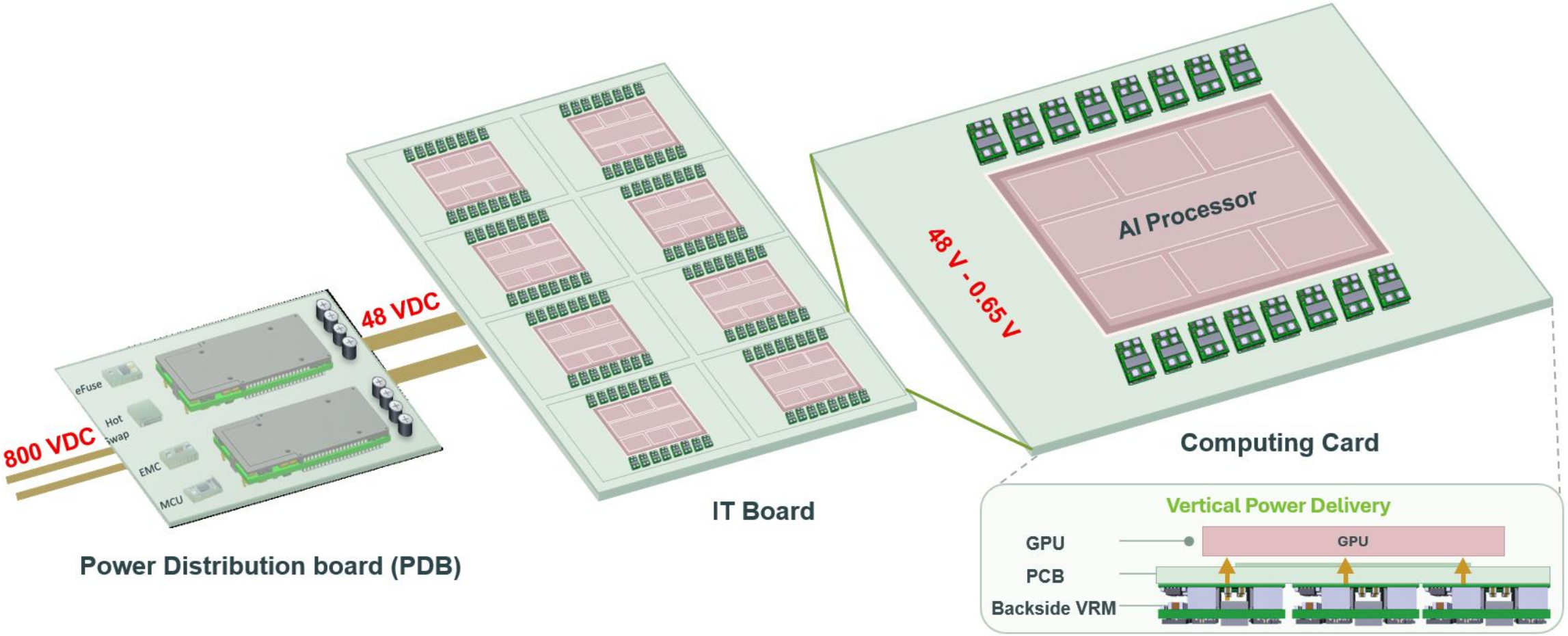
≈1.5 A/mm² current density

2966 W/in³ power density

90.6% peak efficiency

Scaling to Multi-kA Processor Power Delivery

Modular CDR stages scale with standard digital multiphase control



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From Guided Technicians to Lights-Out AI Factories: A Machine-Verifiable Operations Layer

Marco Pritoni, Research Scientist, Lawrence Berkeley National Laboratory

Nauman Rafique, Co-founder, Visum AI

Other contributors: Leonardo Bachega, Co-founder, Visum AI



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Procedures Are Not Linked to Equipment

- AI factory racks are heading to one megawatt, liquid cooling, and high-voltage DC.
- ASHRAE TC 9.9 reports 23 seconds from CDU pump failure to processor throttling.*
- A procedural error during cooling maintenance caused an 11-hour outage that disrupted global trading.†

- Procedures are written for human reading, not machine verification.
- Equipment data sits in engineering drawings, OEM manuals, and SOOs.
- **No semantic graph links procedures to the equipment they operate.**

* ASHRAE TC 9.9, September 2024.

† Uptime Institute, 2026 Annual Outage Analysis.

Outdated Procedures Cause Outages

1 MW

where AI factory racks are heading, liquid cooled, moving to high-voltage DC

Google, OCP EMEA Summit 2025

800+

procedures required for a modern data center

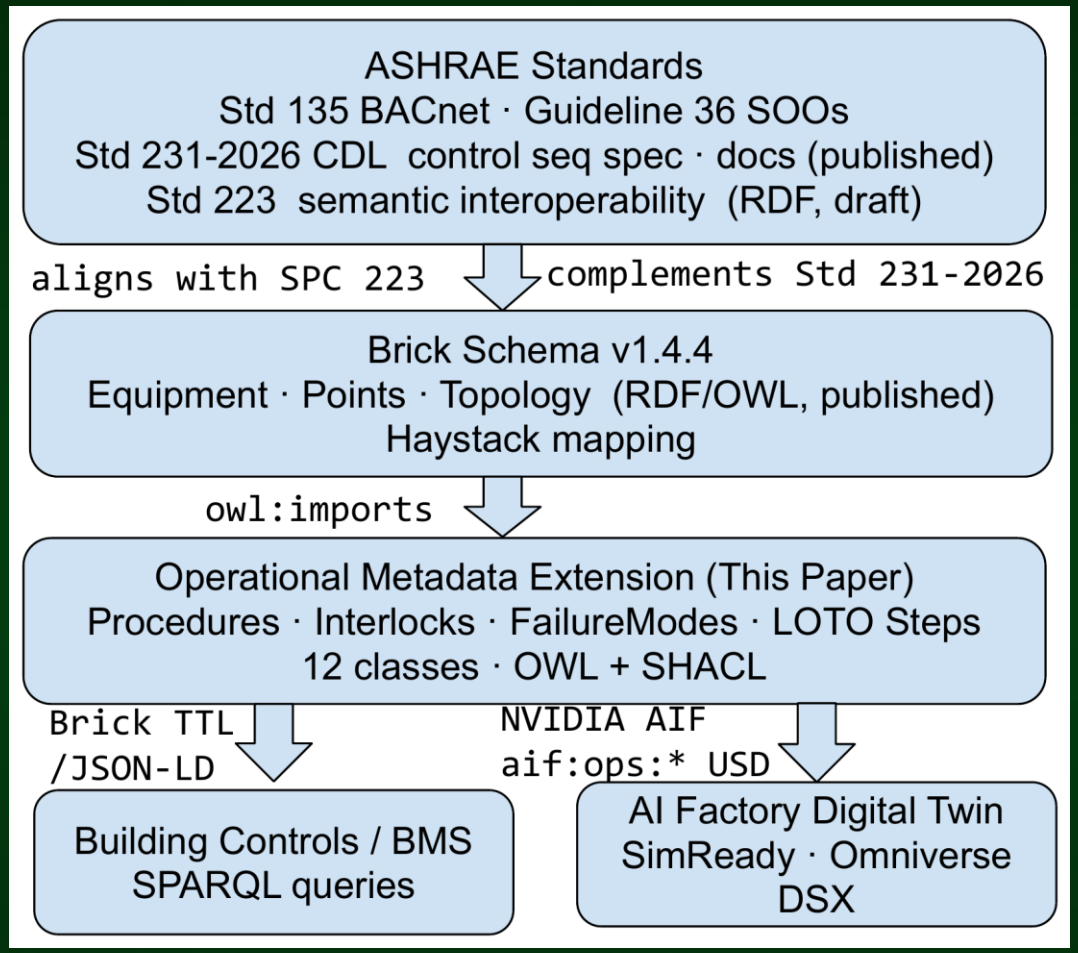
Days

for a senior engineer to write one procedure by hand

Uptime Institute finds the leading human factor in outages is a procedure not followed, most often because it no longer matches the site.

Roughly two thirds of OSHA's FY2025 lockout-tagout citations concerned the written procedure, its training, and its inspection.

One Graph Links Procedures to Equipment

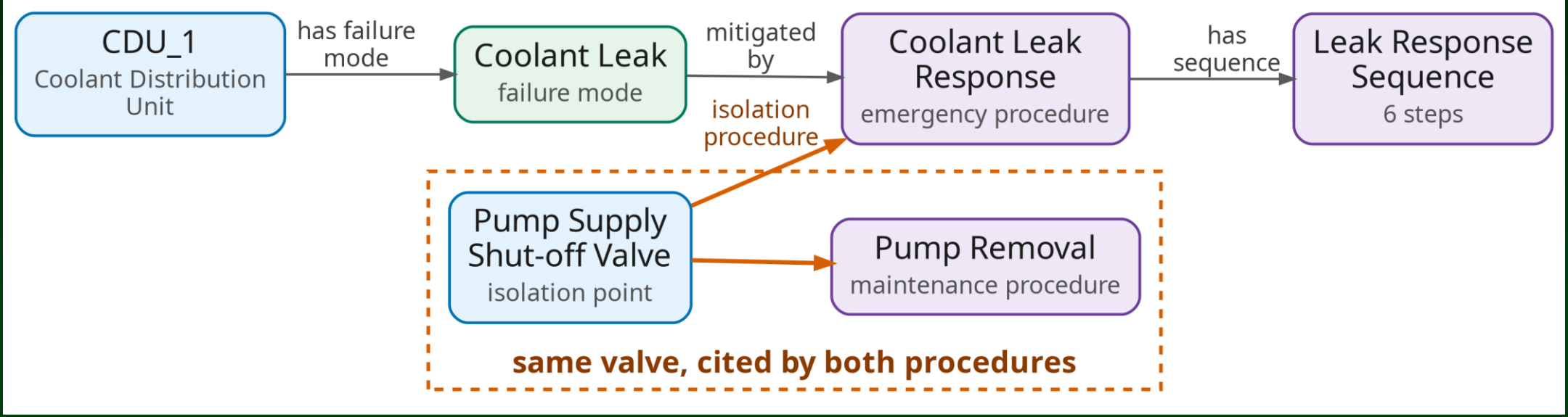


We define a semantic graph that links procedures to the equipment they operate.

We build on the Brick schema, a widely used building ontology, and extend it with operational entity classes for procedures, interlocks, failure modes, isolation points, and lockout-tagout, each backed by a validation template (SHACL shape).

The same graph serializes as Brick TTL for building-controls tooling and as NVIDIA AIF USD attributes for digital-twin platforms.

Validated on the OCP Deschutes CDU Spec



Zero SHACL violations on a full extraction of the Deschutes CDU specification.

Every isolation point cites the section and page of the published spec its requirement comes from.

When a component changes, one query returns every procedure affected by it.

Next Steps

Work with Brick Consortium: a pull request is next. The extension is usable today, since Brick v1.4.4 is already published.

Work with the ASHRAE 223 committee to align with the standard expected in Q4 2026.

Engage a CDU vendor to test the extension at a second facility.

Two to five years until the operational classes land in a published standard, gated on multi-vendor validation.



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